

Ice Detection Strategy

Core idea: We do not claim ice from one signal. We combine independent evidence layers to reduce false positives and produce explainable ice candidates.

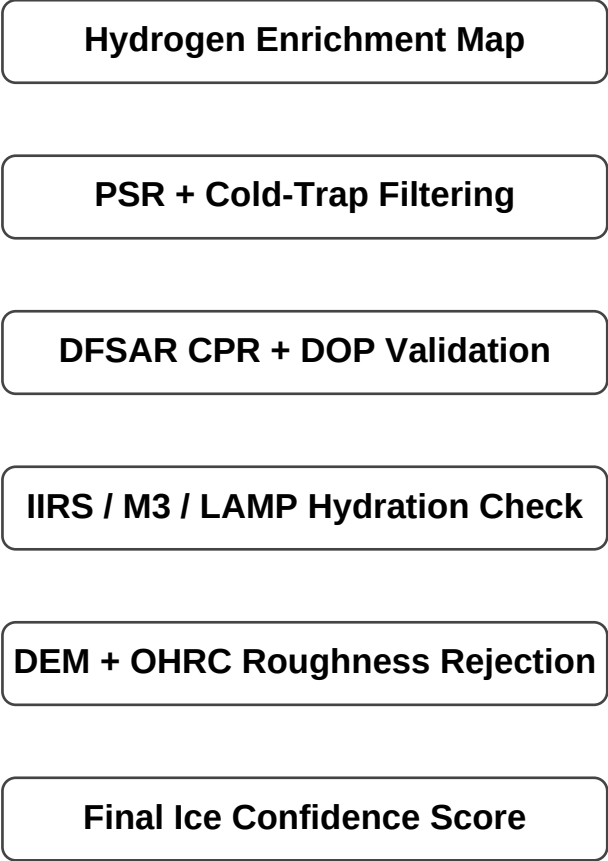
Evidence logic

Layer	Why it matters
Hydrogen	Possible volatile presence
PSR + Thermal	Ice survival condition
Radar CPR/DOP	Buried ice-like scattering
Spectral data	Surface OH/H ₂ O evidence
OHRC + DEM	False-positive rejection

- High CPR being misread as ice when it is only rough rock, boulders, or fresh ejecta.
- Hydrogen-rich regions being treated as water ice without checking cold-trap survival.
- Surface hydration being confused with buried ice.
- Radar geometry effects from slopes and crater walls being mistaken for material evidence.

Speaker line: Hydrogen tells us where to look. Thermal tells us where ice can survive. Radar tells us what may be buried. Optical/topography tells us whether we are being fooled by rocks.

Conservative Ice Confidence Pipeline



Step	Decision role
1. Hydrogen first	Broad inventory of possible volatile-rich zones.
2. PSR + thermal	Rejects places where ice cannot survive.
3. Radar CPR/DOP	Tests for buried volume-scattering behavior.
4. Spectral data	Checks surface or near-surface OH/H2O.
5. OHRC + DEM	Removes rock, slope, boulder, ejecta, and hazard false positives.

Key rule: A site moves forward only when volatile evidence, survival conditions, radar behavior, and terrain context agree.

Mission Output: Explainable Ice Score

Final output: a ranked, explainable target map for landing-site scoring and rover traverse planning.

Conservative scoring formula

$$\text{IceScore} = 0.20H + 0.20\text{PSR} + 0.20T + 0.25R + 0.10\text{Spec} + 0.05\text{Morph} - \text{FalsePositivePenalty}$$

Primary outputs

- Ice confidence map - conservative candidate ranking.
- Per-sensor support map - shows which sensor supports each location.
- Conflict map - highlights sensor disagreement.
- False-positive risk map - boulders, slopes, roughness, ejecta, geometry.
- Candidate zones - input for landing site and rover traverse planning.

Candidate classes

Class	Meaning
Class A	Strong ice candidate: hydrogen + cold trap + CPR/DOP + low false-positive risk.
Class B	Possible buried ice: radar positive, spectral weak, terrain acceptable.
Class C	Surface hydration/frost: spectral/UV positive, radar weak.
Class D	Cold trap only: survivable, but weak ice evidence.
Class E	Likely false positive: radar bright but rocky, rough, warm, or geometry-risky.

The final product is not just an ice map. It is a ranked, explainable priority map showing where a rover should investigate first.